14.

P-value (2-1) = 0.007 < 0.05

CI (2-1) 不包含 0 ⇒ method 1、2 are significant differ

P-value (3-1) = 0.189 > 0.05

CI (3-1) 包含 0 ⇒ method 1.3 X significant differ

P-value (3-2) = 0 < 0.005

CI (3-2) 不包含 0 ⇒ method 3.2 are significant differ

Summary: method 1.3 ⇒ mean X significant differ

且 mean >> method 2

16.

$$\omega = q_\alpha(k, df)\sqrt{\frac{MSE}{n_i}} = q_{0.01}(4, 16) \times \sqrt{\frac{44.25}{5}} \approx 14.91$$

$|\bar{X}_4 - \bar{X}_1| = 6.4 < 14.91$ X

$|\bar{X}_4 - \bar{X}_3| = 17 > 14.91$ ✓

$|\bar{X}_4 - \bar{X}_2| = 33.8 > 14.91$ ✓

$|\bar{X}_1 - \bar{X}_3| = 10.6 < 14.91$ X

$|\bar{X}_1 - \bar{X}_2| = 27.4 > 14.91$ ✓

$|\bar{X}_3 - \bar{X}_2| = 16.8 > 14.91$ ✓

在 $\alpha = 0.01$ 下，state 2 的 mean cost (木材) << other states

state 4 >> state 3，state 2

17. $\bar{X}_L = 608$, $\bar{X}_P = 653$, $\bar{X}_S = 532$

$SST = 74806.67$, $df = 2$ $\quad$ $MST = 37403.33$

$SSE = 225930.00$, $df = 27$ $\quad$ $MSE = 8367.78$

$TSS = 300736.67$, $df = 29$

a. $H_0: \mu_L = \mu_P = \mu_S$, $H_a: \exists(\mu_i \neq \mu_j),\ i \neq j$

$4.47 > F_{2,27,0.05} = 3.35 \Rightarrow$ Reject H_0

b. 95% CI

$$\bar{X}_L - \bar{X}_P \pm t_{df,\frac{\alpha}{2}} \times \sqrt{MSE\left(\frac{1}{n_L} + \frac{1}{n_P}\right)} = -45 \pm 2.052\sqrt{8367.78\left(\frac{1}{10}+\frac{1}{10}\right)}$$

$= (-128.95, 38.95) \Rightarrow$ 包含 0

$\Rightarrow$ X significant differ

c. $$\omega = q_\alpha(k, df) \times \sqrt{\frac{MSE}{n_i}} = 3.51 \times \sqrt{\frac{8367.78}{10}} \approx 101.5$$

$|\bar{X}_P - \bar{X}_L| = 45 < 101.5$ X

$|\bar{X}_L - \bar{X}_S| = 76 < 101.5$ X

$|\bar{X}_P - \bar{X}_S| = 121 > 101.5$ ✓

24.

a. Treatment：不同的超市

Block：不同的商品

b. p-value (Store) = 0 < 0.05 ⇒ Reject H_0

c. p-value (item) = 0 < 0.05 → Reject H_0

Blocking is effective #

25.

a. $q_{0.05}(k, df) = q_{0.05}(4, 21) \approx 3.95$

b. $q_{0.05}(4, 21)\sqrt{\frac{MSE}{b}} = 3.95\sqrt{\frac{0.1658}{8}} \approx 0.569$

c. Ralphs = 4.34 > Vons = 4.1352 Stater Bros = 3.853 > WinCo = 2.681

$|R-W|, |V-W|, |S-W| > 0.569$

$|R-S|, |R-V|, |V-S| < 0.569$

26.

a. treatment : 3 kinds of different mixtures

block : 5 位不同的 investigators

$T_1 = 202$, $T_2 = 212$, $T_3 = 234$

$B_1 = 126$, $B_2 = 113$, $B_3 = 141$, $B_4 = 140$, $B_5 = 128$

$\sum\sum x_{ij} = 648$, $N = 5$, $CM = \frac{648^2}{5} = 27993.6$

Source	df	SS	MS	F
T	2	107.2	53.6	11.05
B	4	176.4	44.1	9.09
E	8	38.8	4.85	
Total	14	322.4		

b. $F_{Tr} = 11.05 > F_{2,8,.001} \approx 8.65$

$\Rightarrow$ Reject H_0, Theare is a significant different in the propellant thruse for the mixtures

c. $F_B = 9.09$, p-value $= 0.005 < 0.01 \Rightarrow$ Reject H_0

d. $\omega = q_{.01}(3,8) \times \sqrt{\frac{4.85}{5}} \approx 5.55$

$|\bar{\mu}_3 - \bar{\mu}_1| = 6.44 > 5.55$ ✓

$|\bar{\mu}_2 - \bar{\mu}_1| = 2 < 5.55$ X

$|\bar{\mu}_3 - \bar{\mu}_2| = 4.4 < 5.55$ X

21.

a. $\hat{y} = a + bx$

$\sum x = 264.9$

$\sum y = 272$

$\sum xy = 14992.61$

$\sum x^2 = 14935.37$

$\bar{x} = 26.49$

$\bar{y} = 27.2$

$b = \frac{\sum xy - n\bar{x}\bar{y}}{\sum x^2 - n\bar{x}^2} = 0.975$

$27.2 = a + 0.975 \times 26.49$, $a \approx 1.38$

$\hat{y} = 1.38 + 0.975x$

b.

The graph 非常合理 $R^2 = 0.9938$

22.

a. The graph said Armspan and Height 之間存在 positive linear relationship

b. $y=x \Rightarrow$ slope $=1$

c. $\Sigma x = 1328$ $\quad \bar{x}=166$ $\quad n=8$

$\Sigma y = 1331$ $\quad \bar{y}=166.375$

$\Sigma xy = 221414$ $\quad \Sigma x^2 = 221016$

$$b = \frac{S_{xy}}{S_{xx}} = \frac{\Sigma xy - n\bar{x}\bar{y}}{\Sigma x^2 - n\bar{x}^2} = 0.824$$

$$a = \bar{y} - 0.824\bar{x} = 29.6$$

$$\Rightarrow \hat{y} = 29.6 + 0.824x$$

d. $\hat{y} = 29.6 + 0.824 \times 157 = 158.968 \approx 159\,cm$

10.

a. indp variable Income (x)

dep variable : BMI (y)

b. $\bar{x} = 40.083$, $\bar{y} = 27$, $\Sigma xy = 6090.65$, $\Sigma x^2 = 12370.54$

$\Sigma y^2 = 4446.22$

$$b = \frac{\Sigma xy - n\bar{x}\bar{y}}{\Sigma x^2 - n\bar{x}^2} = -0.1475, \quad a = 27 - (-0.1475 \times 40.083) \approx 32.91$$

$$\hat{y} = 32.91 - 0.148x$$

c. $SSR = \frac{S_{xy}^2}{S_{xx}} = 59.44$

Source	df	SS	MS	F
Reg	1	59.44	59.44	18.60
E	4	12.78	3.20	
Total	5	72.22		

11.

a. $n=5$, $\bar{x}=298$, $\bar{y}=395.6$ $\Sigma x^2=540100$ $\Sigma xy=653830$

$\Sigma y^2=827504$

$b=\frac{Sxy}{Sxx}=0.67$ $\Rightarrow$ $a=395.6-0.67\times298=195.9$

$\Rightarrow \hat{y}=195.9+0.67x$

b. as the graph shows it provides a relatively good fit to the data points

C. $SSR=\frac{Sxy^2}{Sxx}$

Source	df	SS	MS	F
Reg	1	43146.93	43146.93	69.58
E	3	1860.27	620.09	
Total	4	45007.2		

9.

a. As the graph shows. 隨著年↑. 里程數↑

兩者呈 strongly linear relationship

b. $\bar{x}=7.5$, $\bar{y}=1.04125$, $b=\frac{Sxy}{Sxx}=0.0231$ $a=1.04125-0.0231\times7.5$

$=0.868$

$\hat{y}=0.868+0.0231x$

$H_0: \beta=0$ $H_a: \beta\neq0$

$t^*=\frac{0.0231}{\sqrt{\frac{MSE}{340}}}=22.234>T_{14,0.025}=2.145$

$\Rightarrow$ Reject H_0

c.

Source	df	SS	MS	F
Reg	1	0.181243	0.181243	494.39
E	14	0.005132	0.000367	
Total	15	0.186375		

$494.39 > F_{1,14,0.05} = 4.6 \Rightarrow$ Reject H_0

$t^2 = (22.234)^2 \approx 494.39$

d. $r^2 = \frac{SSR}{SST} = 0.9725$

10.

a. $\hat{y} = 93.617 - 1.251X$

b. $H_0: \beta = 0$, $H_a: \beta \neq 0$

$t^* = -10.471 < -t_{5,0.025} = -2.571$

p-value $= 0 < 0.05 \Rightarrow$ Reject H_0

c. 95% CI $b \pm t_{5,0.025} \times 1.120 = [-1.5595, -0.9425]$

d. $R^2 = 0.9564 = 1096.251 / 49.997$

y variable 有 95.64% 可被解釋